

Deep Dive: 13. Comprehensions

Concept Summary

Comprehensions provide a concise, readable way to create lists, dictionaries, or sets in a single line.

Coding Structure

```
[expression for item in iterable if condition]
```

Example Code

```
nums = [1, 2, 3, 4, 5]
squares = [x*x for x in nums if x % 2 == 0]
print(squares)
```

Code Explanation

- Line 2: This is a List Comprehension. It reads as: 'Give me x*x, for every x in nums, ONLY IF x is even'.
- It acts as a massive time-saver, replacing a bulky 4-line for-loop and if-statement.
- Line 3: Prints [4, 16], which are the squares of 2 and 4.

Practice Problems

1. Use a list comprehension to create a list of all odd numbers between 1 and 20.
2. Create a dictionary comprehension that maps numbers 1-5 to their cubes (e.g., {1: 1, 2: 8}).